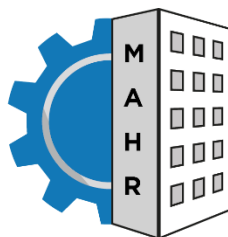


Kingdom of Saudi Arabia
Majmaah University
Ministry of Education
College of Science Al Zulfi
Department of Computer Science
and Information



المملكة العربية السعودية
جامعة المجمعة
وزارة التعليم
كلية العلوم بالزلفي
قسم علوم الحاسب والمعلومات

Managing the Access of Hotel Rooms



Student Affairs System
For College of science Al Zulfi
Department of Computer Science and Information

Graduation Project

Submitted in partial fulfillment of the requirements for the award of
Bachelor degree of Majmaah University
(Semester 1, 2021-22)

Submitted by:

Sami AlKhunayni 381100005

Mohammad AlAtiwi 381101600

Under the supervision of:

Dr. Abdullah Alourani

Abstract

Hotel management by the traditional way is along difficult process which involves manual work and consumption of time. The current project is designed to manage the access of hotel rooms using a barcode-based system called Managing the Access of Hotel Rooms (MAHR). This system issues a barcode for each reservation and helps the hotel management and guests to communicate with each other using the chat and to provide and request the services through the app. In addition, this environmental-friendly system may reduce the need to the printed cards. The main feature of this project is the facilitation of hotel management services to the staff and guests.

Acknowledgement

We would like to express our gratitude and appreciation for our supervisor Dr. Abdullah Alourani for his invaluable help, which have contributed to the improvement of this project.

The deepest and warmest thanks are expressed to all people who have helped and supported us during the process of developing the project.

**MAJMAAH UNIVERSITY,
COLLEGE OF SCIENCE AL ZULFI,
DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION**

(CERTIFICATE BY STUDENT)

This is to certify that the project titled **“Managing the Access of Hotel Rooms”** submitted by us (**Sami AlKhunayni 381100005; Mohammad AlAtiwi 381101600**) under the supervision of **Dr. Abdullah Alourani** for award of Bachelor degree of Majmaah University carried out during the Semester 1, 2021-22 embodies our original work.

Signature in full: **Sami AlKhunayni**

Name in block letters: **Sami AlKhunayni**

Student ID: **381100005**

Signature in full: **Mohammad AlAtiwi**

Name in block letters: **Mohammad AlAtiwi**

Student ID: **381101600**

Date: 2/12/2021

Table of Contents

Chapter One: Introduction	6
1.1 Problem Definition.....	6
1.1.1 Goals	6
1.1.2 Scope.....	6
1.1.3 Objectives	7
1.1.4 Critical success factors.....	7
1.2 General rules	7
Chapter Two: System Analysis and Specification	8
2.1 Introduction.....	8
2.2 Data Modeling	8
2.2.1 The Entity of Relationships Diagram (ERD) of Project	8
2.2.2 Use case	10
2.2.3 Sequence Diagram	13
2.2.4 UML Class Database Diagram	21
Chapter Three: System design.....	22
3.1 Description of procedures and functions	22
3.1.1 Function	22
3.1.2 Operation.....	22
3.1.3 Process	23
3.2 Relation database schema	23
3.2.1 Tables and Attributes	24
3.2.2 Database schema	27
3.2.3 Code	28
Chapter Four: Implementation and Testing	35
4.1 Layouts.....	35
4.1.1 Guest	35
4.1.2 Service Provider.....	38
4.1.3 Admin	40
4.2 Conclusion	43

Table of Figures

Figure 1.ERD	10
Figure 2.Use case (Admin)	11
Figure 3.Use case (Guest)	12
Figure 4.Use case (Service Provider).....	13
Figure 5.Admin Login	14
Figure 6.Story Administration (A).....	15
Figure 7.Story Administration (B).....	16
Figure 8.Update Profile By Admin.	16
Figure 9.Login OTP PIN (Guest).....	17
Figure 10.Notification (Guest).....	17
Figure 11.Access Barcode Room (Guest).....	18
Figure 12.Story (Guest)	18
Figure 13.Choose Service	19
Figure 14.Update Profile By Guest.....	19
Figure 15.Story Service Provider.....	20
Figure 16.UML Class Database Diagram	21
Figure 17.Database Schema.....	27
Figure 18.SignIn (Guest)	35
Figure 19.Guest Homepage	36
Figure 20.Room Cleaning.....	36
Figure 21.Profile (Guest)	37
Figure 22.Chat (Guest).....	37
Figure 23.Notification (Guest).....	38
Figure 24.Service provider interface.....	39
Figure 25.Service delivery	39
Figure 26.Log in (Admin).....	40
Figure 27.Service status	40
Figure 28.Chat.....	41
Figure 29.Data Management (Reservations)	41
Figure 30.Data Management (Guest).....	42
Figure 31.Data Management (Employees)	42

Chapter One: Introduction

Currently, a large proportion of operating hotels use card lock systems to access the rooms. Usually, the hotel management replaces the card when the card is lost or damaged by another card. The method of receiving requests from the guest and the hotel management is by calling or speaking to a member of room service. The traditional way consumes time and effort as well as a large number of printed cards. This project aims to replace the cards with a barcode using a system to facilitate the process of managing the room.

1.1 Problem Definition

The current project addresses the problem of excessive consumption of card printing. In addition, sometimes there are delays in executing requests due to lack of attention or pressure in the requests, especially at peak times. From another perspective, traditional card locking systems are often more expensive than barcode systems. Another disadvantage of using traditional systems deals with their programming speed, which takes more time than setting up a barcode.

1.1.1 Goals

The main goal of the project is to create an application that helps to manage hotel rooms technically. Moreover, the use of smart devices facilitate procedures and save the time by providing guest data in the app using a barcode-based system to access to the room. Another goal which may be regarded an important especially in crisis time such as COVID-19 is to reduce contact between people and find a contactless way to protect themselves.

1.1.2 Scope

The scope of this project is to provide guest data in the app by a barcode-based system. Then, the hotel management continues with the guests to remind them of the beginning and end of the reservation and also in case the food is processed. Furthermore, the guest reached out to the hotel

management to clean the room or to make observations and evaluate the service by giving their feedback.

1.1.3 Objectives

The project aims to achieve three objectives in order to enhance the workflow of hotel management by the replacement of door cards printing with a barcode-based system, providing comfort to all, and writing feedback to the hotel.

1.1.4 Critical success factors

The success of the implementation of the project depends on some critical factors such as the technical skills to Create the MAHR App by Dart and DB by Firebase, good communication skills to market the app to hotels.

1.2 General rules

This system will be developed to work on a barcode-based app and web browser. Thus, it is assumed that the users will have access to a computer or a mobile phone and to the internet.

Chapter Two: System Analysis and Specification

2.1 Introduction

This section demonstrates how the system works and represents the logical relationships of entity sets to explain the workflow of the system.

2.2 Data Modeling

It is the creation of types of graphics that explain the mechanism of the project or system through texts and symbols.

2.2.1 The Entity of Relationships Diagram (ERD) of Project

An entity-relationship diagram (ERD) explains how the connections of entity groups are gathered in a database. An entity here refers to an item, a part of data. An entity set is defined as a group of similar entities which have features that determine their properties. An ER diagram (ERD) “models the data structure of a reality in terms of entities, relationships, and attributes” (Shoval,2004:217).

2.2.1.1 Description of Entities

- I. Employee:** Manage the guest profile, Manage the services, And Communicate with the Guest By chat.
- II. Guest:** The guest gives his personal data to the reservations and create a barcode for him to enter the room and enables him to request services.
- III. Barcode:** Generates a barcode for the guest and uses it to enter the room.
- IV. Service:** Types of services in the application such as cleaning the room.
- V. Room:** It is data for the guest to know the location of the room.
- VI. Reservation:** It is to register the guest's complete data to create a custom barcode for him.

2.2.1.2 Description of Relations

2.2.1.2.1 Employee – Guest relationship

This is a many-to-many relationship which in each employee or more than one employee can chat with one guest or more than one guest and vice versa.

2.2.1.2.2 Employee – Reservation relationship

This is many to many relationship which in each employee or more than one employee can manage to book one or more reservations and vice versa.

2.2.1.2.3 Employee – Barcode relationship

This is many to many relationship which in each employee or more than one employee can manage a barcode for one or more barcodes and vice versa.

2.2.1.2.4 Guest – Barcode relationship

This is one to one relationship which in each guest can access to one barcode.

2.2.1.2.5 Reservation – Guest relationship

This is many to one relationship in which reservation or more than one reservation requests data for each guest.

2.2.1.2.6 Reservation – Barcode relationship

This is one to one relationship which in each reservation generates one barcode.

2.2.1.2.7 Reservation – Room relationship

This is one to many relationships in which each reservation is assigned to each room one or more rooms.

2.2.1.2.8 Room – Barcode relationship

This is one to one relationship which in each room is given one barcode.

2.2.1.2.9 Room – Services relationship

This is many to many relationships in which each room or more than one room works with each service or more than one service.

2.2.1.2.10 Service – Barcode relationship

This is many to one relationship which in each or more services have access to one barcode.

2.2.1.3 Drawing ERD

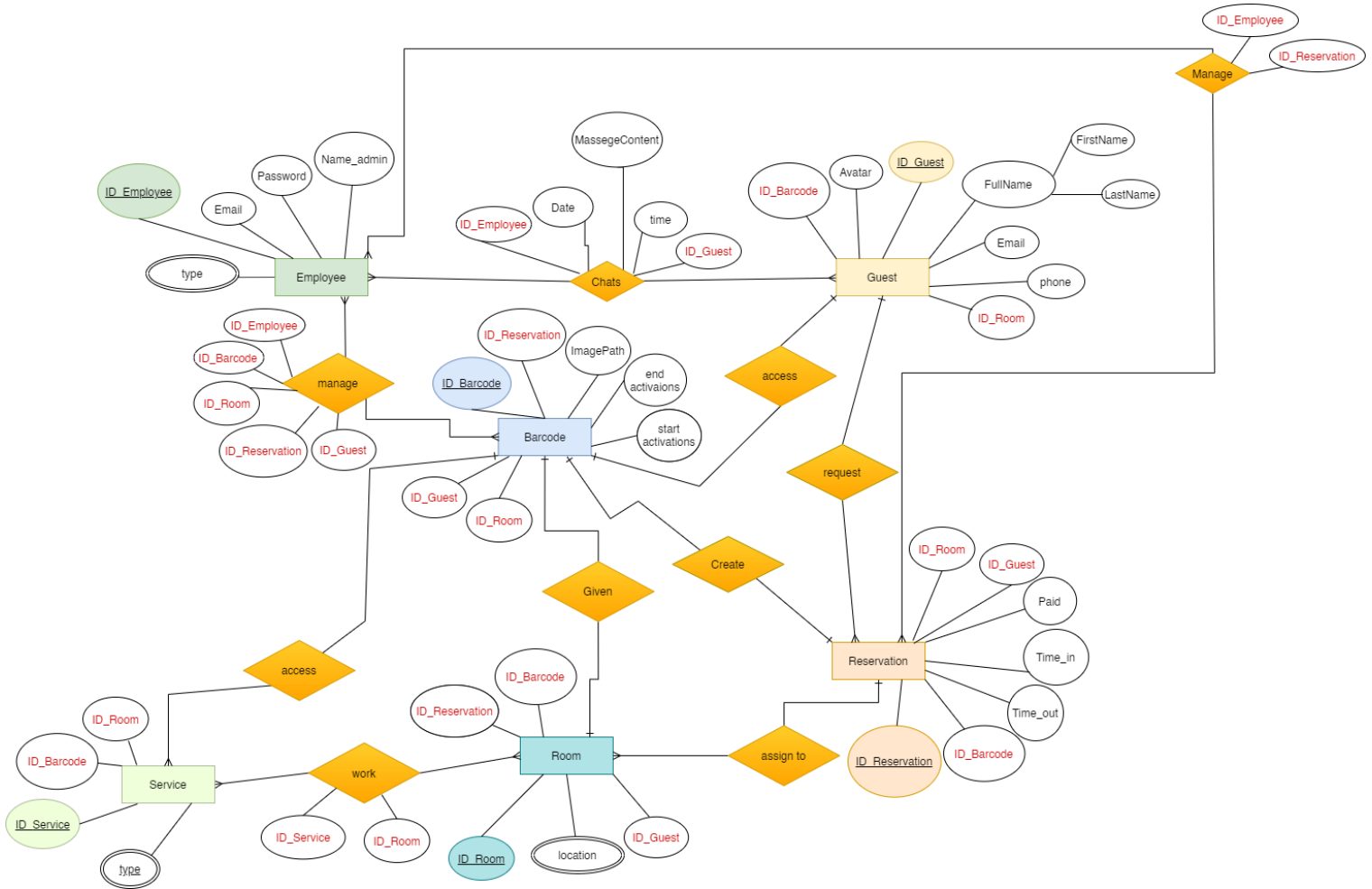


Figure 1.ERD

2.2.2 Use case

Use case is an important and useful requirement to explain how the system works. A use case is defined as a description of a series of events which result in a system to perform a task (Bittner and Spence, 2003). According to Bittner and Spence (2003), the use case is useful for many people who have a relationship with the system such as customers, managers, analysts, developers, testers, technical writers, and user-experience designers.

In figure 2, the tasks of admin in the app are represented below.

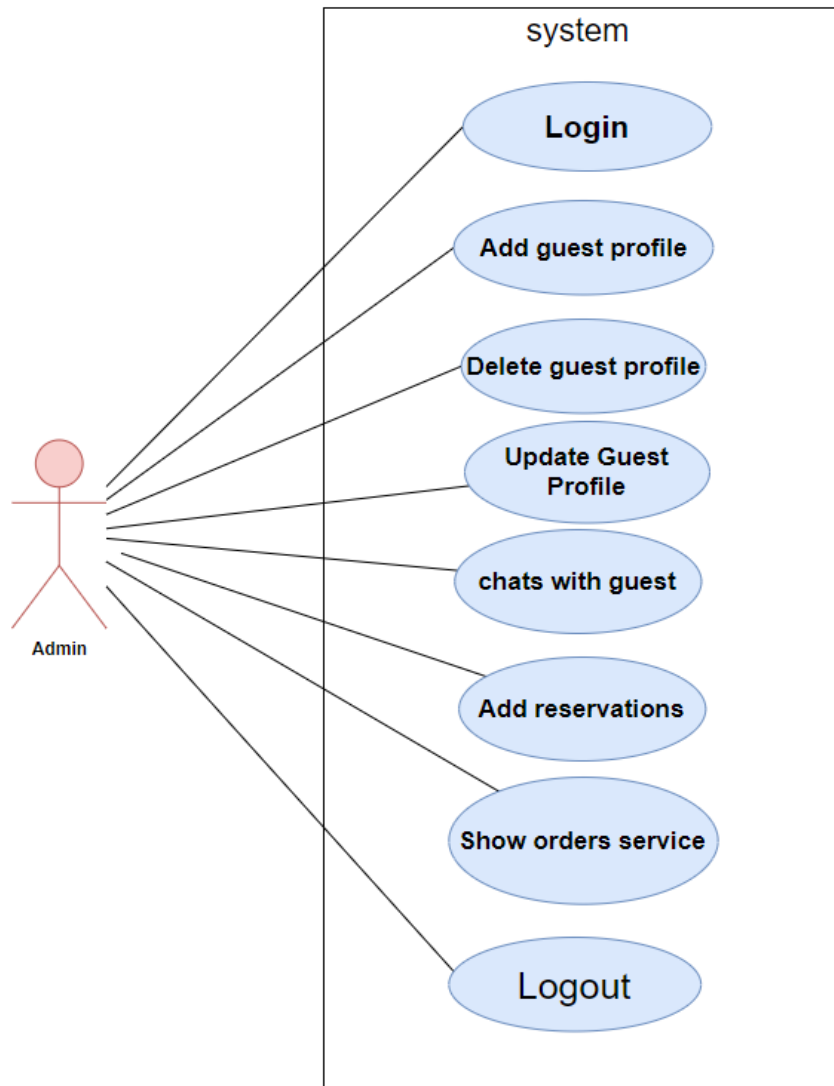


Figure 2. Use case (Admin)

In figure 3, the tasks of guest in the app are represented below.

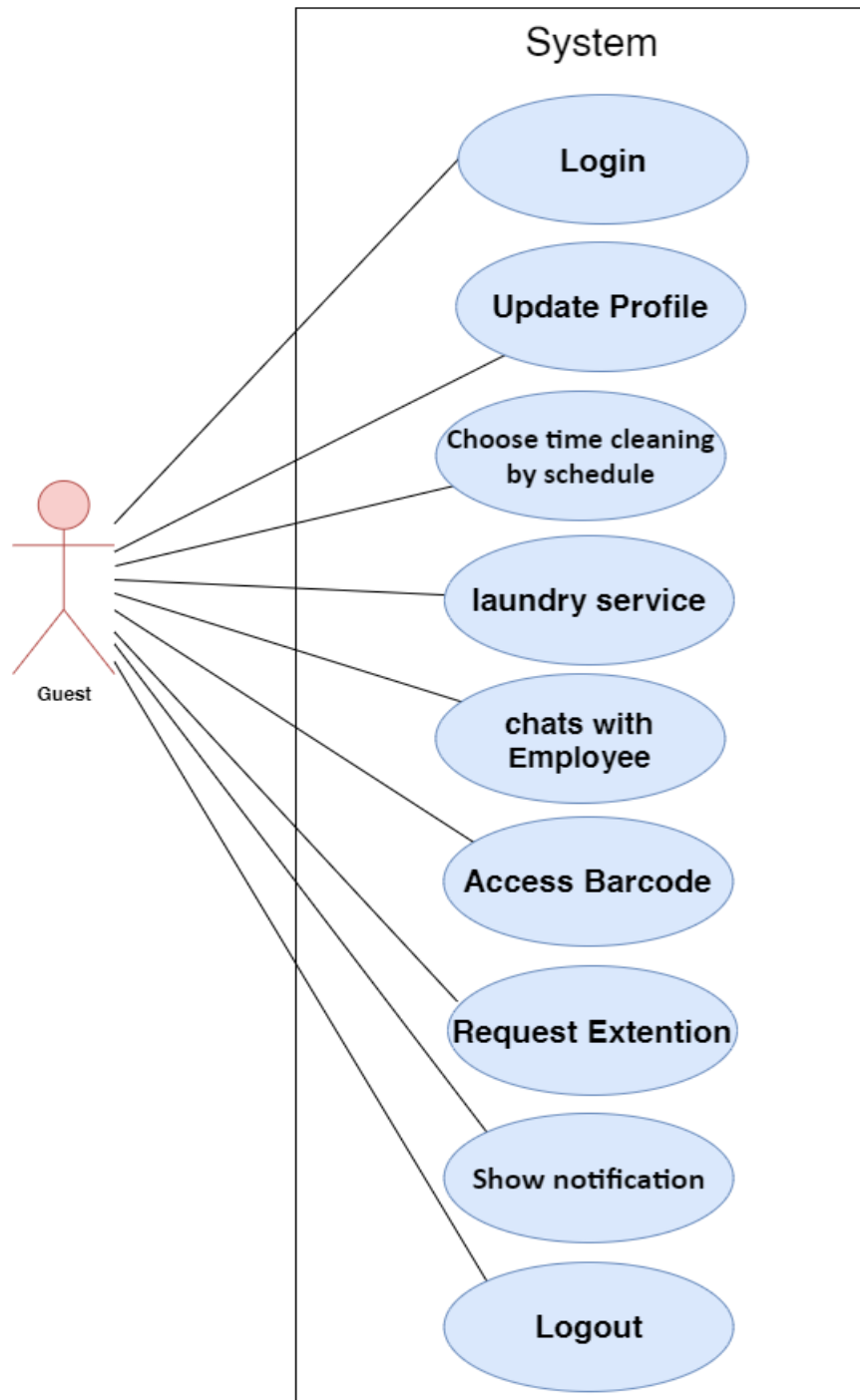


Figure 3. Use case (Guest)

In figure 4, the tasks of a service provider in the app are represented below.

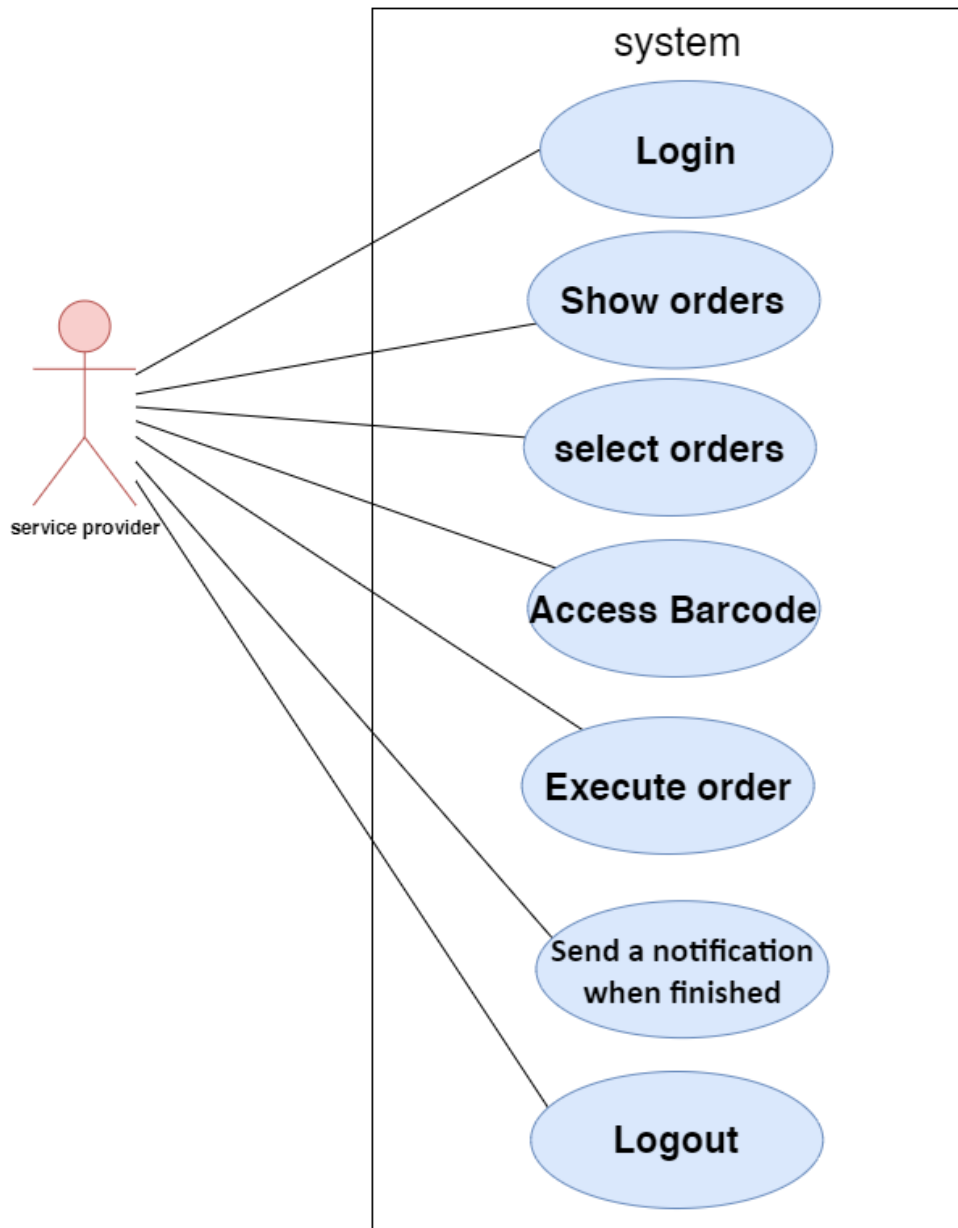


Figure 4. Use case (Service Provider)

2.2.3 Sequence Diagram

Sequence Diagrams are representative diagrams that explain the workflow of a system operations. Alvin et al (2021:31) define sequence diagram as “visual representations of object

interactions in a system and can provide valuable information for program comprehension, debugging, maintenance, and software archeology”.

2.2.3.1. Admin

Admin have four sequence diagrams as follows:

I. Log In

At first, the admin logs in on the home page and verifies the login data by sending the data to the databases by accepting or rejecting the entry. In the case of acceptance, it goes to the contents of the page, and in case of rejection, it goes to the forget password page to renew the password, and on the forgotten password page, the email is entered and the databases are checked for the validity of the email entered. If it is proven, send a message to an email to specify the password. If the email is not correct, it will return to the forgotten password page and eventually log out as it is shown in figure 5.

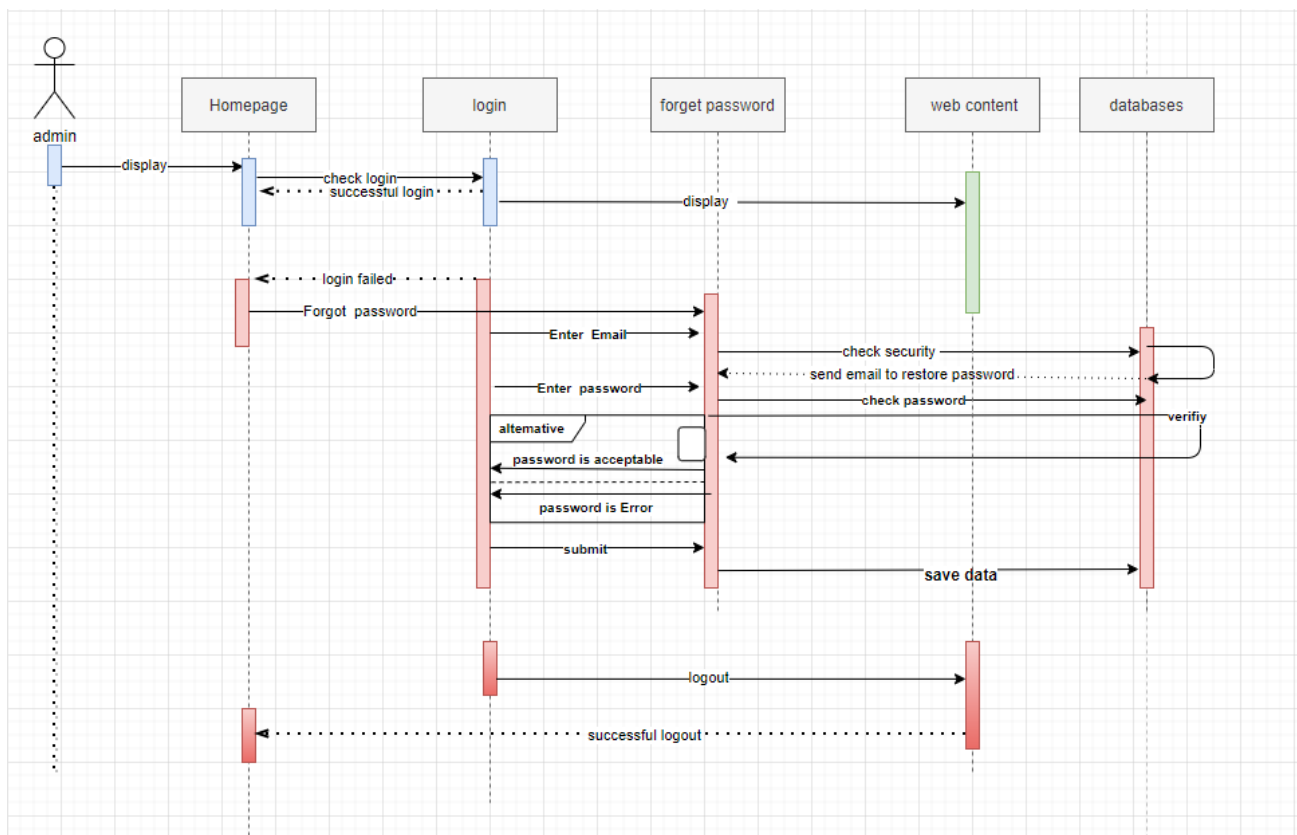


Figure 5.Admin Login

II. Story Administration A.

Login then reviews and manage reservations (to verify payment and extension) then check the status of services (ensure that all services required for guests have been completed) and communicate via chat with guests (ensure that there are new requests via chat such as cleaning clothes) and finally sign out as it is shown in figure 6.

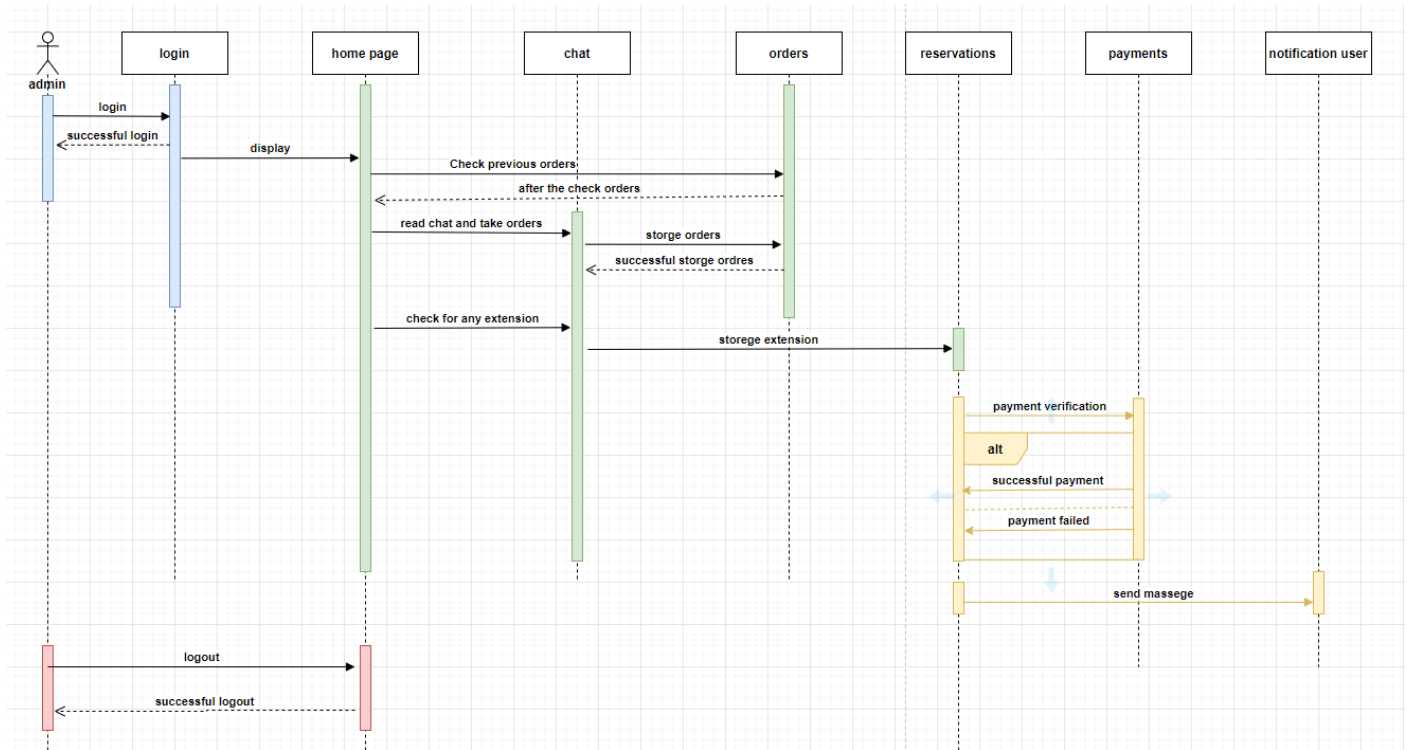


Figure 6. Story Administration (A)

III. Story Administration B.

Login then check the status of services (ensure that all services required for guests have been completed) and communicate via chat with guests (ensure that there are new requests via chat such as cleaning clothes) and finally sign out as it is shown in figure 7.

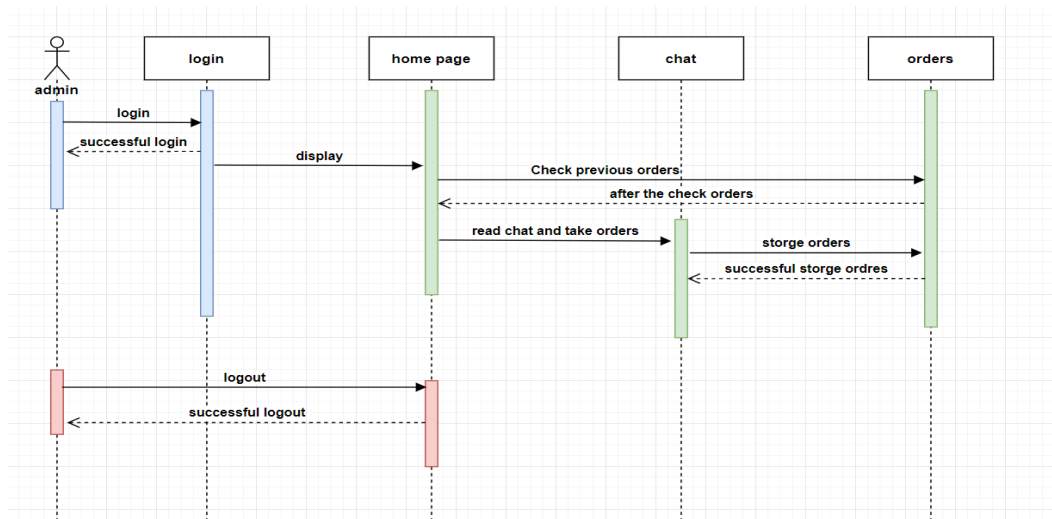


Figure 7.Story Administration (B)

IV. Update Profile by Admin.

update an Admin profile and make sure not to use a phone number or email that was used on another profile in the databases as it is shown in figure 8.

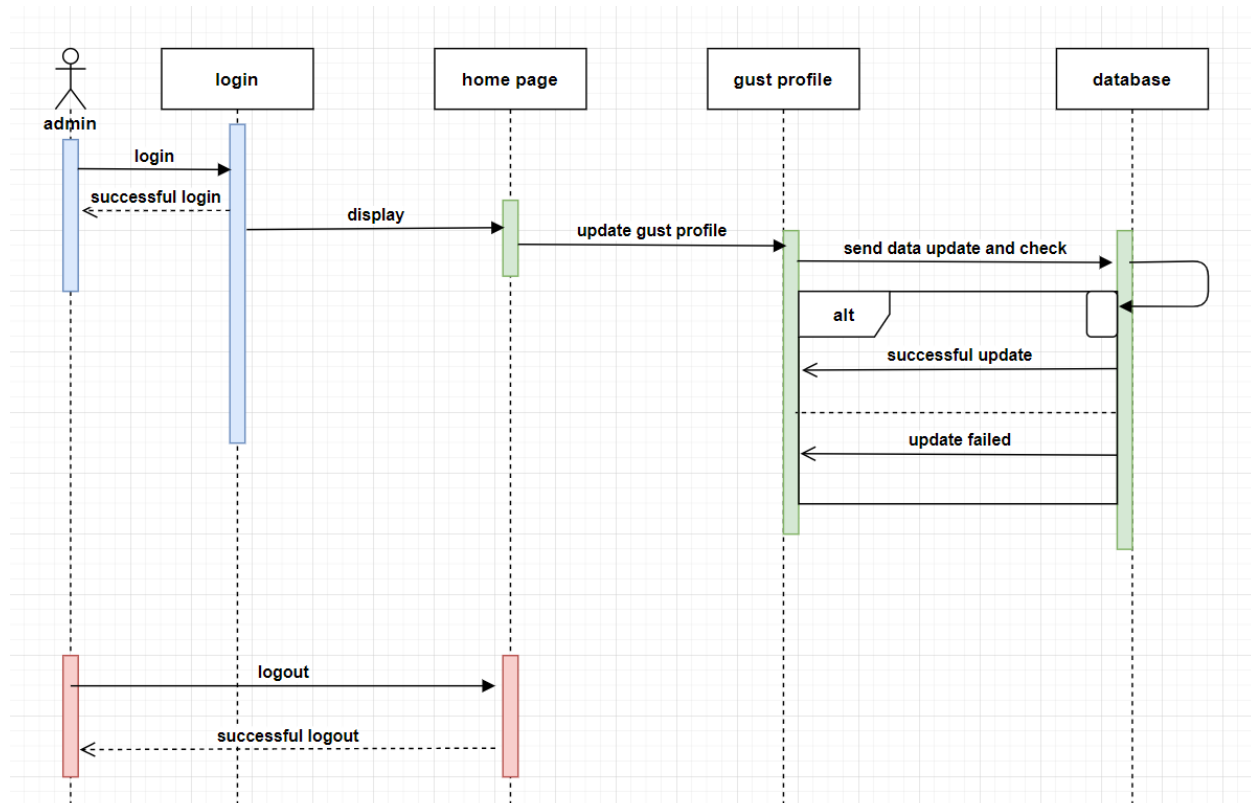


Figure 8.Update Profile By Admin.

2.2.4 Guest

I. Login OTP PIN.

At first, the guest enters the phone number on the login page, and the phone number is verified, and if it is proven, he sends a message containing a verification number on the phone number (PIN), and then enters the PIN verification number, verify the number in the databases, log in and go to the home page as it is shown in figure 9.

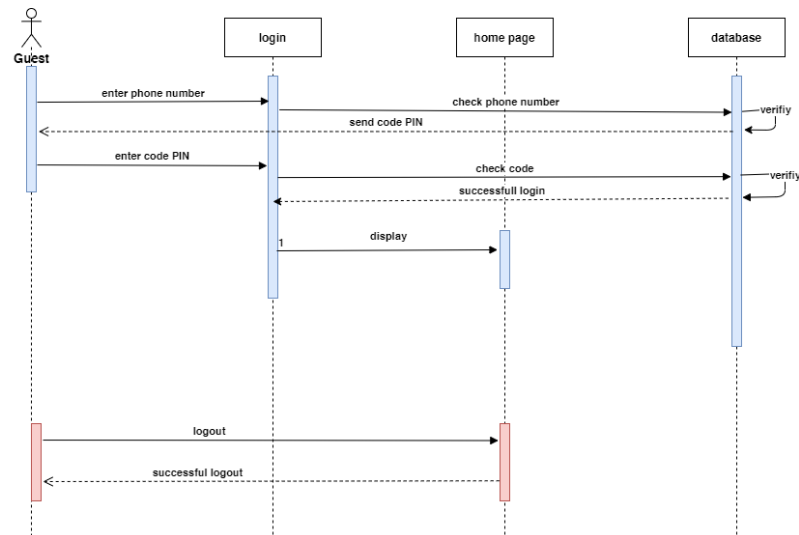


Figure 9.Login OTP PIN (Guest)

II. Notification Guest

Figure 10 explains how to send alerts to the guest.

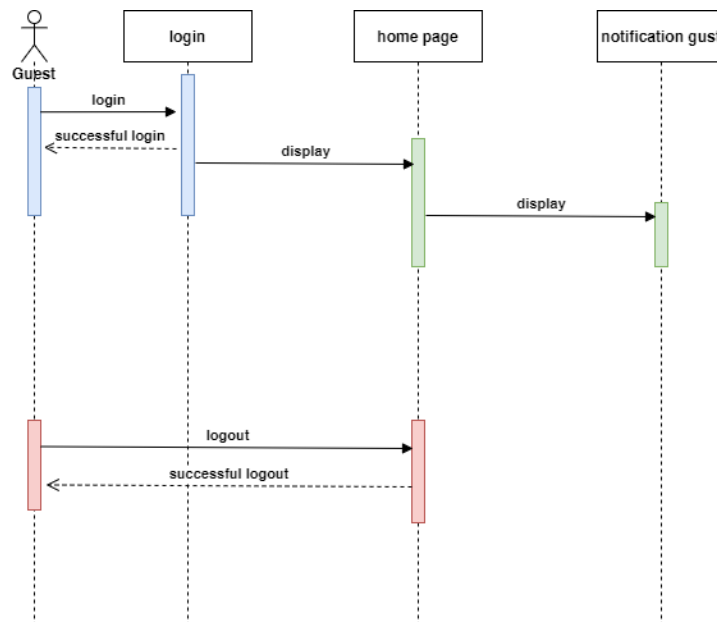


Figure 10.Notification (Guest)

III. Access Barcode Room

Figure 11 explain how to use a barcode to access rooms.

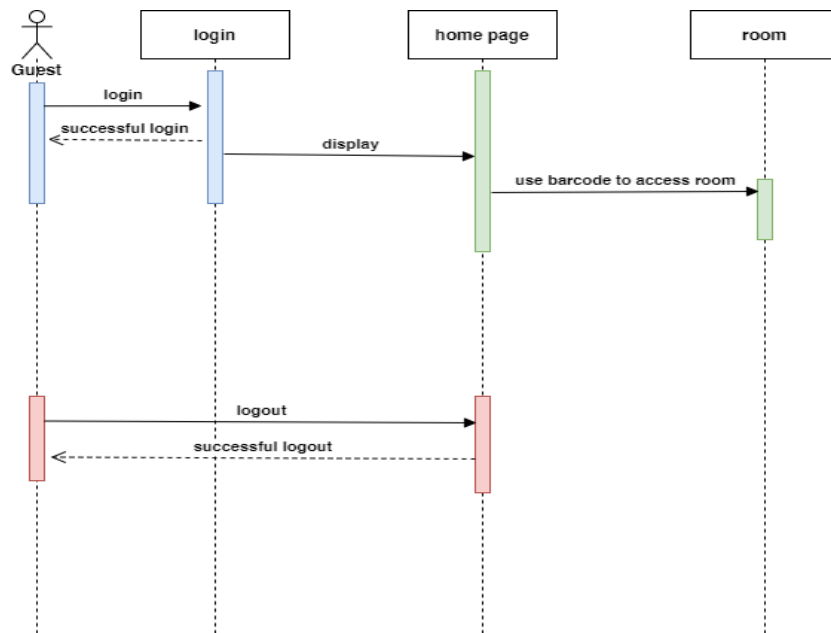


Figure 11. Access Barcode Room (Guest)

IV. Story Guest

Log in and show the Barcode for Access to enter the room then select service and can chat with Admin and can show the notifications then log out as it is shown in figure 12.

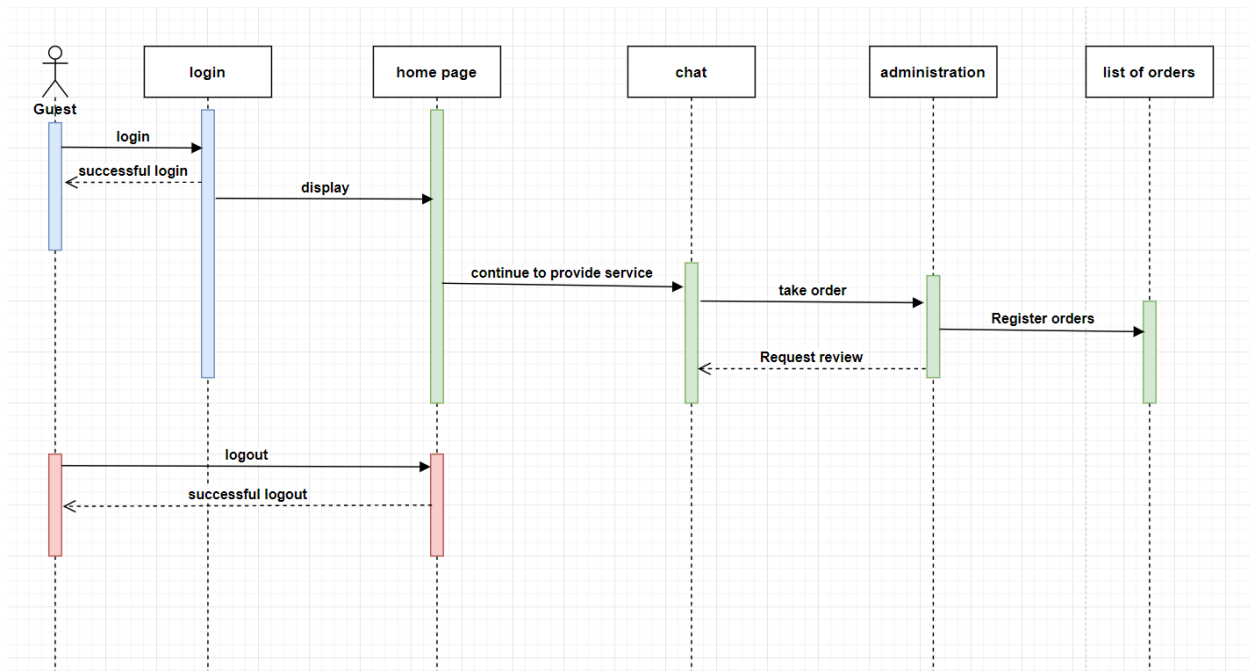


Figure 12. Story (Guest)

V. Choose Service

At first, log in, go to the services, and choose the service and the appropriate time, and then it is sent to the orders in the databases, and in the end, log out as it is shown in figure 13.

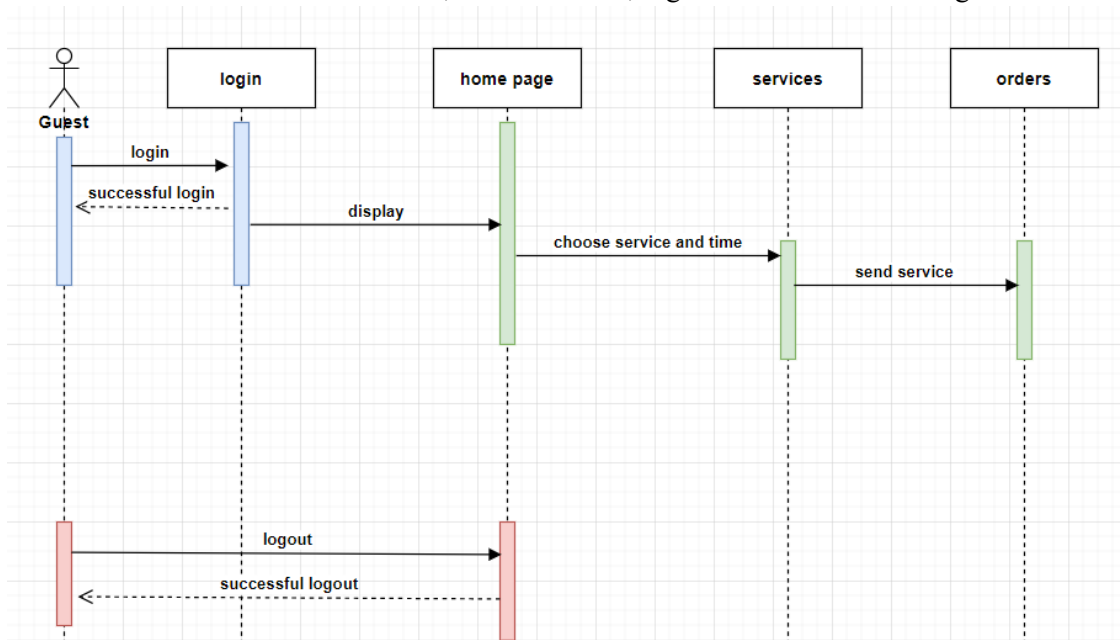


Figure 13. Choose Service

VI. Update Profile by Guest.

Update a guest's profile and make sure not to use a phone number or email that was used on another profile in the databases as it is shown in figure 14.

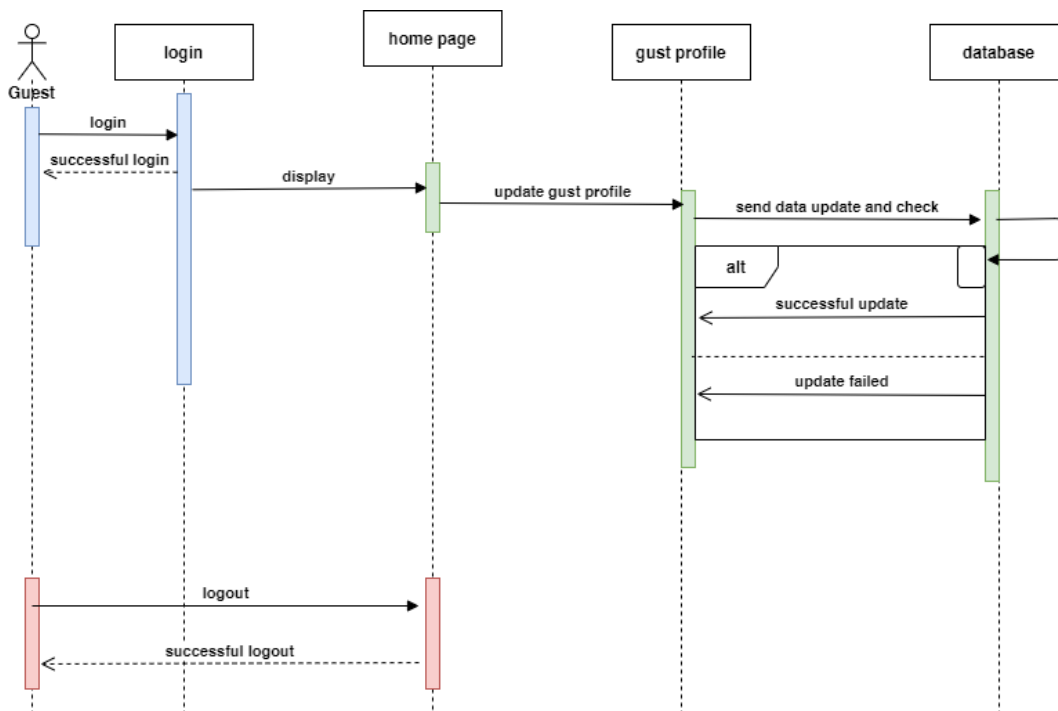


Figure 14. Update Profile By Guest.

2.2.5 Service Provider

I. Story Service Provider.

Login then the list of requests then choose the request and it will be executed then the status of the request (completed or not) and finally logging out as it is shown in figure 15.

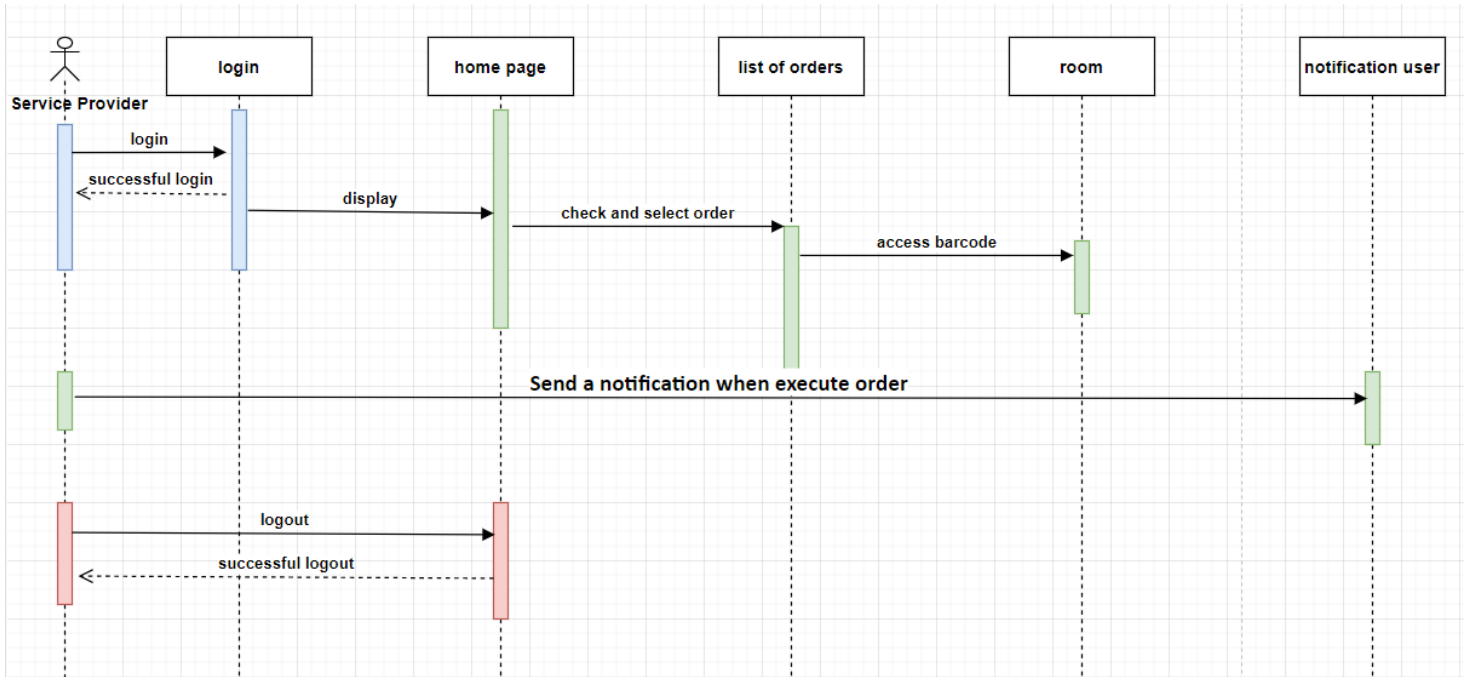


Figure 15. Story Service Provider.

2.2.4 UML Class Database Diagram

The class diagram shows the building blocks of any object-orientated system. Class diagrams depict a static view of the model, or part of the model, describing what attributes and behavior it has rather than detailing the methods for achieving operations. Class diagrams are most useful in illustrating relationships between classes and interfaces as it is represented by figure 16.

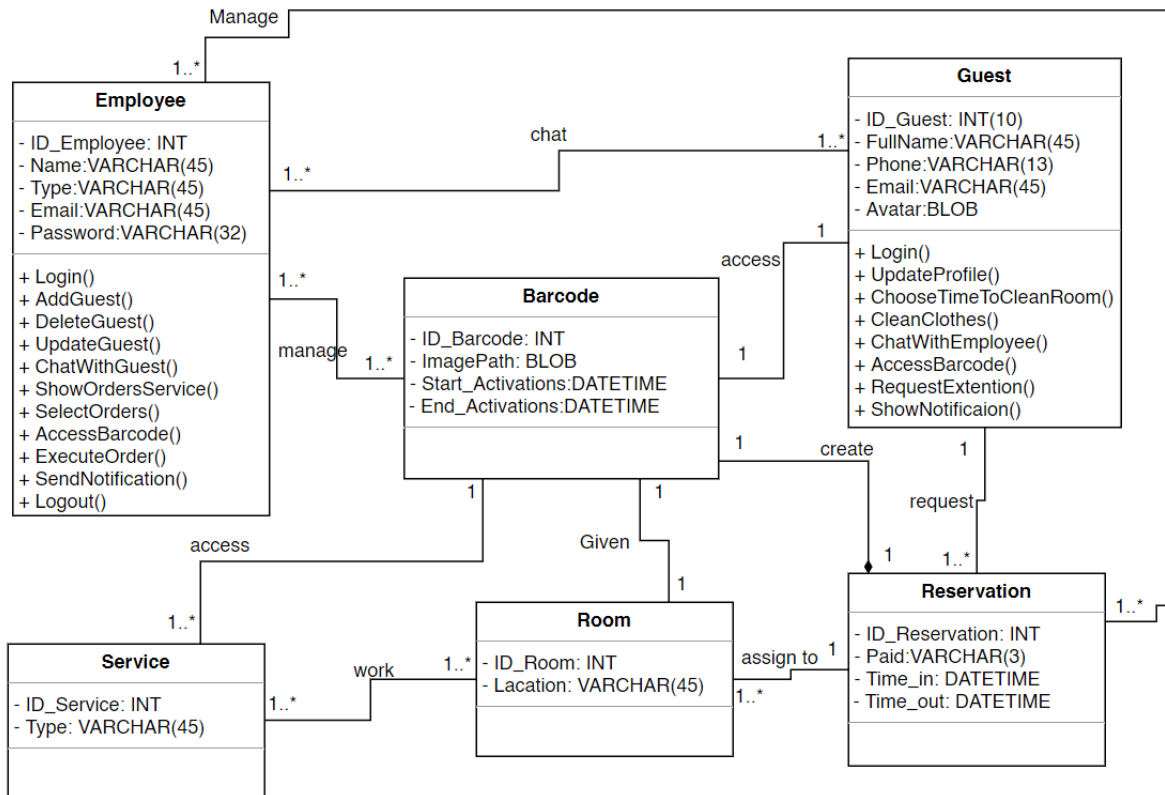


Figure 16.UML Class Database Diagram

Chapter Three: System design

3.1 Description of procedures and functions

3.1.1 Function

3.1.1.1 Admin

- I. Add guest profile.
- II. Delete guest profile.
- III. Update Guest Profile.
- IV. Chat with a guest.
- V. Show orders service.

3.1.1.2 Guest

- I. Update profile
- II. Choose time cleaning by schedule.
- III. Choose service
- IV. Chat with an employee.
- V. Access to the barcode.
- VI. Show notification.

3.1.1.3 Service Provider

- I. show orders.
- II. select orders.
- III. Access to the barcode.
- IV. Execute order.
- V. Send a notification when finished.

3.1.2 Operation

3.1.2.1 Admin

- I. Manage the guest profile.
- II. Manage the services.
- III. Communicate with Guest By chat.

3.1.2.2 Guest

- I. Manage the profile (Update profile, show Barcode, Notifications, Select services).
- II. Communicate with Admin By chat.

3.1.2.3 Service Provider

- I. Manage orders.

3.1.3 Process

3.1.3.1 Admin

- I. Login then review and manage reservations (to verify payment and extension) then check the status of services (ensure that all services required for guests have been completed) and communicate via chat with guests (ensure that there are new requests via chat such as cleaning clothes) and finally sign out.
- II. Login then check the status of services (ensure that all services required for guests have been completed) and communicate via chat with guests (ensure that there are new requests via chat such as cleaning clothes) and finally sign out.

3.1.3.2 Guest

Log in and show the Barcode for Access to enter the room then select service and can chat with Admin and can show the notifications then log out.

3.1.3.3 Service Provider

Login then the list of requests then choose the request and it will be executed then the status of the request (completed or not) and finally logging out.

3.2 Relation database schema

Relationships are explained such as one to one, one to many, or many to many also an Attribute and its types are clarified.

3.2.1 Tables and Attributes

Employee		
Attribute	Type	Notice
Id_Employee	Integer	Primary Key
Name	Varchar (45)	
Type	Varchar (45)	
Email	Varchar (45)	Unique
Password	Varchar (32)	

Employee Chat_Guest		
Attribute	Type	Notice
Id_Chat	Integer	Primary Key
Date_And Time	Datetime	
Massegecontent	Varchar (45)	
Id_Employee	Integer	Foreign Key from Table Employee
Id_Guest	Integer	Foreign Key from Table Guest

Guest		
Attribute	Type	Notice
Id_Guest	Integer	Primary Key
FullName	Varchar (45)	
Phone	Integer	Unique
Email	Varchar (45)	Unique
Avatar	BLOB	
Id_Barcode	Integer	Foreign Key from Table Barcode
Id_Room	Integer	Foreign Key from Table Room

Employee Manage Reservation		
Attribute	Type	Notice
Id_Employee	Integer	Foreign Key from Table Employee
Id_Reservation	Integer	Foreign Key from Table Reservation

Reservation		
Attribute	Type	Notice
Id Reservation	Integer	Primary Key
Paid	Varchar (3)	
Time in	DateTime	
Time Out	DateTime	
Id_Guest	Integer	Foreign Key from Table Guest
Id_Barcode	Integer	Foreign Key from Table Barcode
Id_Room	Integer	Foreign Key from Table Room

Room		
Attribute	Type	Notice
Id Room	Integer	Primary Key
Location	Varchar (45)	
Id_Reservation	Integer	Foreign Key from Table Reservation
Id_Guest	Integer	Foreign Key from Table Guest
Id_Barcode	Integer	Foreign Key from Table Barcode
Id_Room	Integer	Foreign Key from Table Room

Service work Room		
Attribute	Type	Notice
Id_Service	Integer	Foreign Key from Table Service
Id_Room	Integer	Foreign Key from Table Room

Service		
Attribute	Type	Notice
Id_Service	Integer	Primary Key
Type	Varchar (45)	
Id_Barcode	Integer	Foreign Key from Table Barcode
Id_Room	Integer	Foreign Key from Table Room

Employee_Manage_Barcode		
Attribute	Type	Notice
Id_Employee	Integer	Foreign Key from Table Employee
Id_Barcode	Integer	Foreign Key from Table Barcode
Id_Room	Integer	Foreign Key from Table Room
Id_Reservation	Integer	Foreign Key from Table Reservation
Id_Guest	Integer	Foreign Key from Table Guest

Barcode		
Attribute	Type	Notice
Id_Barcode	Integer	Primary Key
ImagePath	BLOB	Unique
Start_Activations	DateTime	
End_Activations	DateTime	
Id_Reservation	Integer	Foreign Key from Table Reservation
Id_Guest	Integer	Foreign Key from Table Guest
Id_Room	Integer	Foreign Key from Table Room

3.2.2 Database schema

In figure 17, there are 10 tables and their relationships, for example, one-to-one relationship, one to many relationship and many to many as well as the types of attributes such as integer, varchar, and blob, and also the constraints on the attributes such as the primary key.

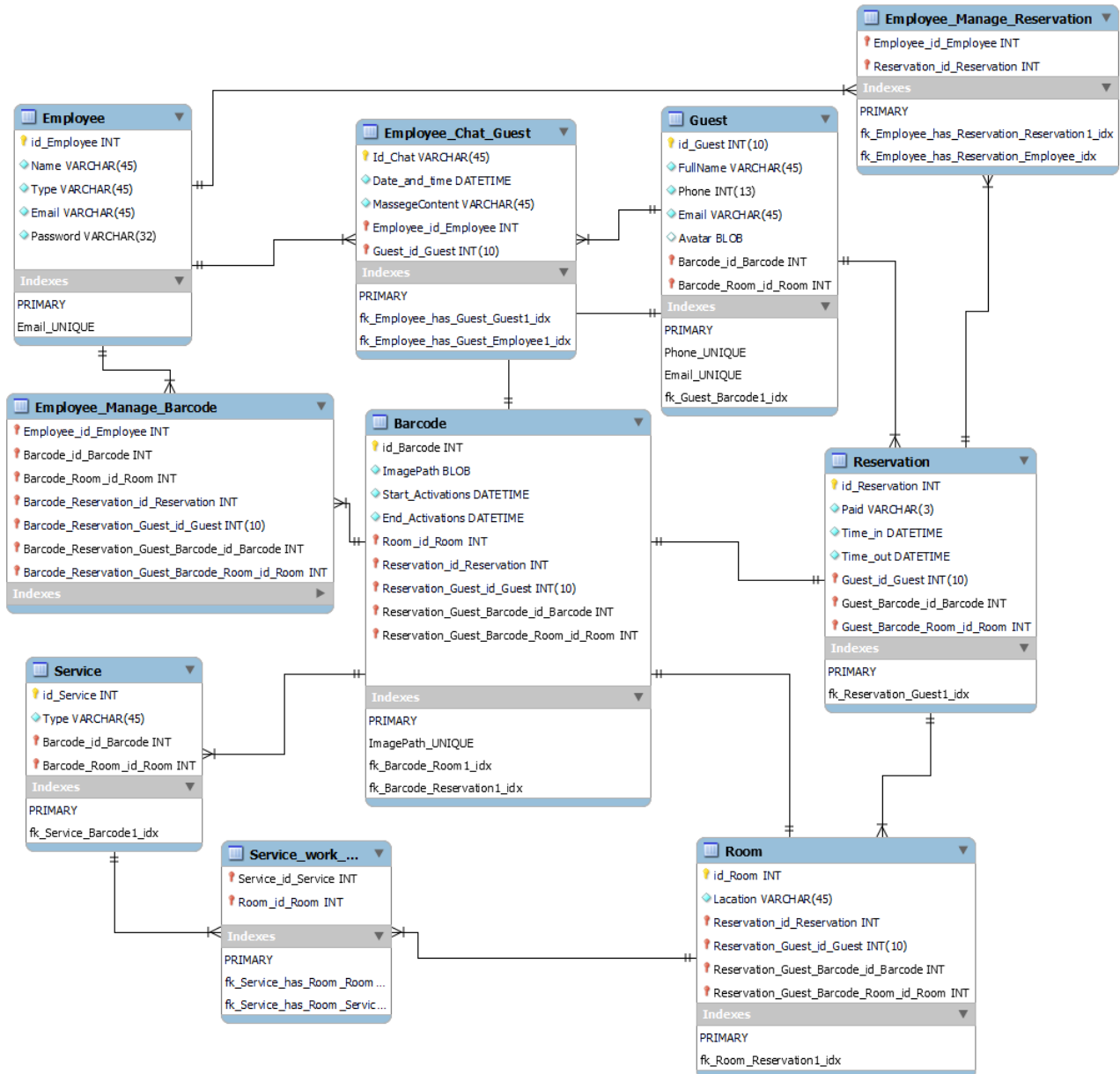


Figure 17. Database Schema

3.2.3 Code

The code is written to create the project database which clarify the types of attributes such as integer, varchar, and datetime. These attributes are described by constraints such as the primary key, the foreign key, and the unique key (Raval and Trivedi, 2013). In this section, the system codes will be analyzed.

```
CREATE SCHEMA IF NOT EXISTS `DbMAHR` DEFAULT CHARACTER SET utf8 ;  
USE `DbMAHR` ;
```

3.2.3.1 Employee

```
CREATE TABLE IF NOT EXISTS `DbMAHR`.`Employee` (  
  `id_Employee` INT NOT NULL,  
  `Name` VARCHAR(45) NOT NULL,  
  `Type` VARCHAR(45) NOT NULL,  
  `Email` VARCHAR(45) NOT NULL,  
  `Password` VARCHAR(32) NOT NULL,  
  PRIMARY KEY (`id_Employee`),  
  UNIQUE INDEX `Email_UNIQUE` (`Email` ASC) VISIBLE)  
ENGINE = InnoDB;
```

3.2.3.2 Room

```
CREATE TABLE IF NOT EXISTS `DbMAHR`.`Room` (  
  `id_Room` INT NOT NULL,  
  `Lacation` VARCHAR(45) NOT NULL,  
  `Reservation_id_Reservation` INT NOT NULL,  
  `Reservation_Guest_id_Guest` INT(10) NOT NULL,  
  `Reservation_Guest_Barcode_id_Barcode` INT NOT NULL,  
  `Reservation_Guest_Barcode_Room_id_Room` INT NOT NULL,  
  PRIMARY KEY (`id_Room`, `Reservation_id_Reservation`, `Reservation_Guest_id_Guest`,  
  `Reservation_Guest_Barcode_id_Barcode`, `Reservation_Guest_Barcode_Room_id_Room`),  
  INDEX `fk_Room_Reservation1_idx` (`Reservation_id_Reservation` ASC, `Reservation_Guest_id_Guest` ASC,  
  `Reservation_Guest_Barcode_id_Barcode` ASC, `Reservation_Guest_Barcode_Room_id_Room` ASC) VISIBLE,  
  CONSTRAINT `fk_Room_Reservation1`
```

```

FOREIGN KEY ('Reservation_id_Reservation', 'Reservation_Guest_id_Guest',
'Reservation_Guest_Barcode_id_Barcode', 'Reservation_Guest_Barcode_Room_id_Room')
REFERENCES `DbMAHR`.`Reservation` ('id_Reservation', 'Guest_id_Guest', 'Guest_Barcode_id_Barcode',
'Guest_Barcode_Room_id_Room'))
ENGINE = InnoDB;

```

3.2.3.3 Barcode

```

CREATE TABLE IF NOT EXISTS `DbMAHR`.`Barcode` (
  `id_Barcode` INT NOT NULL,
  `ImagePath` BLOB NOT NULL,
  `Start_Activations` DATETIME NOT NULL,
  `End_Activations` DATETIME NOT NULL,
  `Room_id_Room` INT NOT NULL,
  `Reservation_id_Reservation` INT NOT NULL,
  `Reservation_Guest_id_Guest` INT(10) NOT NULL,
  `Reservation_Guest_Barcode_id_Barcode` INT NOT NULL,
  `Reservation_Guest_Barcode_Room_id_Room` INT NOT NULL,
  PRIMARY KEY ('id_Barcode', 'Room_id_Room', 'Reservation_id_Reservation', 'Reservation_Guest_id_Guest',
'Reservation_Guest_Barcode_id_Barcode', 'Reservation_Guest_Barcode_Room_id_Room'),
  UNIQUE INDEX `ImagePath_UNIQUE` ('ImagePath' ASC) VISIBLE,
  UNIQUE INDEX `End_Activations_UNIQUE` ('End_Activations' ASC) VISIBLE,
  INDEX `fk_Barcode_Room1_idx` ('Room_id_Room' ASC) VISIBLE,
  INDEX `fk_Barcode_Reservation1_idx` ('Reservation_id_Reservation' ASC, 'Reservation_Guest_id_Guest'
ASC, 'Reservation_Guest_Barcode_id_Barcode' ASC, 'Reservation_Guest_Barcode_Room_id_Room' ASC)
VISIBLE,
  CONSTRAINT `fk_Barcode_Room1`
    FOREIGN KEY ('Room_id_Room')
      REFERENCES `DbMAHR`.`Room` ('id_Room')
    ON DELETE NO ACTION
    ON UPDATE NO ACTION,
  CONSTRAINT `fk_Barcode_Reservation1`
    FOREIGN KEY ('Reservation_id_Reservation', 'Reservation_Guest_id_Guest',
'Reservation_Guest_Barcode_id_Barcode', 'Reservation_Guest_Barcode_Room_id_Room')

```

```
REFERENCES `DbMAHR`.`Reservation` (`id_Reservation`, `Guest_id_Guest`, `Guest_Barcode_id_Barcode`,  
`Guest_Barcode_Room_id_Room`)
```

```
ON DELETE NO ACTION
```

```
ON UPDATE NO ACTION)
```

```
ENGINE = InnoDB;
```

3.2.3.4 Service

```
CREATE TABLE IF NOT EXISTS `DbMAHR`.`Service` (
```

```
`id_Service` INT NOT NULL,
```

```
`Type` VARCHAR(45) NOT NULL,
```

```
`Barcode_id_Barcode` INT NOT NULL,
```

```
`Barcode_Room_id_Room` INT NOT NULL,
```

```
PRIMARY KEY (`id_Service`, `Barcode_id_Barcode`, `Barcode_Room_id_Room`),
```

```
INDEX `fk_Service_Barcode1_idx` (`Barcode_id_Barcode` ASC, `Barcode_Room_id_Room` ASC) VISIBLE,
```

```
CONSTRAINT `fk_Service_Barcode1`
```

```
FOREIGN KEY (`Barcode_id_Barcode`, `Barcode_Room_id_Room`)
```

```
REFERENCES `DbMAHR`.`Barcode` (`id_Barcode`, `Room_id_Room`)
```

```
ON DELETE NO ACTION
```

```
ON UPDATE NO ACTION)
```

```
ENGINE = InnoDB;
```

3.2.3.5 Reservation

```
CREATE TABLE IF NOT EXISTS `DbMAHR`.`Reservation` (
```

```
`id_Reservation` INT NOT NULL,
```

```
`Paid` VARCHAR(3) NOT NULL,
```

```
`Time_in` DATETIME NOT NULL,
```

```
`Time_out` DATETIME NOT NULL,
```

```
`Guest_id_Guest` INT(10) NOT NULL,
```

```
`Guest_Barcode_id_Barcode` INT NOT NULL,
```

```
`Guest_Barcode_Room_id_Room` INT NOT NULL,
```

```

PRIMARY KEY (`id_Reservation`, `Guest_id_Guest`, `Guest_Barcode_id_Barcode`,
`Guest_Barcode_Room_id_Room`),

INDEX `fk_Reservation_Guest1_idx` (`Guest_id_Guest` ASC, `Guest_Barcode_id_Barcode` ASC,
`Guest_Barcode_Room_id_Room` ASC) VISIBLE,

CONSTRAINT `fk_Reservation_Guest1`

FOREIGN KEY (`Guest_id_Guest`, `Guest_Barcode_id_Barcode`, `Guest_Barcode_Room_id_Room`)

REFERENCES `DbMAHR`.`Guest` (`id_Guest`, `Barcode_id_Barcode`, `Barcode_Room_id_Room`)

ON DELETE NO ACTION

ON UPDATE NO ACTION)

ENGINE = InnoDB;

```

3.2.3.6 Guest

```

CREATE TABLE IF NOT EXISTS `DbMAHR`.`Guest` (

`id_Guest` INT(10) NOT NULL,

`FullName` VARCHAR(45) NOT NULL,

`Phone` INT(13) NOT NULL,

`Email` VARCHAR(45) NOT NULL,

`Avatar` BLOB NULL,

`Barcode_id_Barcode` INT NOT NULL,

`Barcode_Room_id_Room` INT NOT NULL,

PRIMARY KEY (`id_Guest`, `Barcode_id_Barcode`, `Barcode_Room_id_Room`),

UNIQUE INDEX `Phone_UNIQUE` (`Phone` ASC) VISIBLE,

UNIQUE INDEX `Email_UNIQUE` (`Email` ASC) VISIBLE,

INDEX `fk_Guest_Barcode1_idx` (`Barcode_id_Barcode` ASC, `Barcode_Room_id_Room` ASC) VISIBLE,

CONSTRAINT `fk_Guest_Barcode1`

FOREIGN KEY (`Barcode_id_Barcode`, `Barcode_Room_id_Room`)

REFERENCES `DbMAHR`.`Barcode` (`id_Barcode`, `Room_id_Room`)

ON DELETE NO ACTION

ON UPDATE NO ACTION)

ENGINE = InnoDB;

```


3.2.3.7 Employee_Manage_Reservation

```
CREATE TABLE IF NOT EXISTS `DbMAHR`.`Employee_Manage_Reservation` (  
  `Employee_id_Employee` INT NOT NULL,  
  `Reservation_id_Reservation` INT NOT NULL,  
  PRIMARY KEY (`Employee_id_Employee`, `Reservation_id_Reservation`),  
  INDEX `fk_Employee_has_Reservation_Reservation1_idx` (`Reservation_id_Reservation` ASC) VISIBLE,  
  INDEX `fk_Employee_has_Reservation_Employee_idx` (`Employee_id_Employee` ASC) VISIBLE,  
  CONSTRAINT `fk_Employee_has_Reservation_Employee`  
    FOREIGN KEY (`Employee_id_Employee`)  
      REFERENCES `DbMAHR`.`Employee` (`id_Employee`)  
    ON DELETE NO ACTION  
    ON UPDATE NO ACTION,  
  CONSTRAINT `fk_Employee_has_Reservation_Reservation1`  
    FOREIGN KEY (`Reservation_id_Reservation`)  
      REFERENCES `DbMAHR`.`Reservation` (`id_Reservation`)  
    ON DELETE NO ACTION  
    ON UPDATE NO ACTION)  
ENGINE = InnoDB;
```

3.2.3.8 Employee_Chat_Guest

```
CREATE TABLE IF NOT EXISTS `DbMAHR`.`Employee_Chat_Guest` (  
  `Id_Chat` VARCHAR(45) NOT NULL,  
  `Date_and_time` DATETIME NOT NULL,  
  `MassegeContent` VARCHAR(45) NOT NULL,  
  `Employee_id_Employee` INT NOT NULL,  
  `Guest_id_Guest` INT(10) NOT NULL,  
  PRIMARY KEY (`Id_Chat`, `Employee_id_Employee`, `Guest_id_Guest`),  
  INDEX `fk_Employee_has_Guest_Guest1_idx` (`Guest_id_Guest` ASC) VISIBLE,  
  INDEX `fk_Employee_has_Guest_Employee1_idx` (`Employee_id_Employee` ASC) VISIBLE,
```

```

CONSTRAINT `fk_Employee_has_Guest_Employee1`
FOREIGN KEY (`Employee_id_Employee`)
REFERENCES `DbMAHR`.`Employee` (`id_Employee`)
ON DELETE NO ACTION
ON UPDATE NO ACTION,
CONSTRAINT `fk_Employee_has_Guest_Guest1`
FOREIGN KEY (`Guest_id_Guest`)
REFERENCES `DbMAHR`.`Guest` (`id_Guest`)
ON DELETE NO ACTION
ON UPDATE NO ACTION)
ENGINE = InnoDB;

```

3.2.3.9 Service_work_Room

```

CREATE TABLE IF NOT EXISTS `DbMAHR`.`Service_work_Room` (
`Service_id_Service` INT NOT NULL,
`Room_id_Room` INT NOT NULL,
PRIMARY KEY (`Service_id_Service`, `Room_id_Room`),
INDEX `fk_Service_has_Room_Room1_idx` (`Room_id_Room` ASC) VISIBLE,
INDEX `fk_Service_has_Room_Service1_idx` (`Service_id_Service` ASC) VISIBLE,
CONSTRAINT `fk_Service_has_Room_Service1`
FOREIGN KEY (`Service_id_Service`)
REFERENCES `DbMAHR`.`Service` (`id_Service`)
ON DELETE NO ACTION
ON UPDATE NO ACTION,
CONSTRAINT `fk_Service_has_Room_Room1`
FOREIGN KEY (`Room_id_Room`)
REFERENCES `DbMAHR`.`Room` (`id_Room`)
ON DELETE NO ACTION
ON UPDATE NO ACTION)
ENGINE = InnoDB;

```

3.2.3.10 Employee_Manage_Barcode

```
CREATE TABLE IF NOT EXISTS `DbMAHR`.`Employee_Manage_Barcode` (  
  `Employee_id_Employee` INT NOT NULL,  
  `Barcode_id_Barcode` INT NOT NULL,  
  `Barcode_Room_id_Room` INT NOT NULL,  
  `Barcode_Reservation_id_Reservation` INT NOT NULL,  
  `Barcode_Reservation_Guest_id_Guest` INT(10) NOT NULL,  
  `Barcode_Reservation_Guest_Barcode_id_Barcode` INT NOT NULL,  
  `Barcode_Reservation_Guest_Barcode_Room_id_Room` INT NOT NULL,  
  PRIMARY KEY (`Employee_id_Employee`, `Barcode_id_Barcode`, `Barcode_Room_id_Room`,  
  `Barcode_Reservation_id_Reservation`, `Barcode_Reservation_Guest_id_Guest`,  
  `Barcode_Reservation_Guest_Barcode_id_Barcode`, `Barcode_Reservation_Guest_Barcode_Room_id_Room`),  
  INDEX `fk_Employee_has_Barcode_Barcode1_idx` (`Barcode_id_Barcode` ASC, `Barcode_Room_id_Room`  
  ASC, `Barcode_Reservation_id_Reservation` ASC, `Barcode_Reservation_Guest_id_Guest` ASC,  
  `Barcode_Reservation_Guest_Barcode_id_Barcode` ASC,  
  `Barcode_Reservation_Guest_Barcode_Room_id_Room` ASC) VISIBLE,  
  INDEX `fk_Employee_has_Barcode_Employee1_idx` (`Employee_id_Employee` ASC) VISIBLE,  
  CONSTRAINT `fk_Employee_has_Barcode_Employee1`  
    FOREIGN KEY (`Employee_id_Employee`)  
    REFERENCES `DbMAHR`.`Employee` (`id_Employee`)  
    ON DELETE NO ACTION  
    ON UPDATE NO ACTION,  
  CONSTRAINT `fk_Employee_has_Barcode_Barcode1`  
    FOREIGN KEY (`Barcode_id_Barcode`, `Barcode_Room_id_Room`, `Barcode_Reservation_id_Reservation`,  
  `Barcode_Reservation_Guest_id_Guest`, `Barcode_Reservation_Guest_Barcode_id_Barcode`,  
  `Barcode_Reservation_Guest_Barcode_Room_id_Room`)  
    REFERENCES `DbMAHR`.`Barcode` (`id_Barcode`, `Room_id_Room`, `Reservation_id_Reservation`,  
  `Reservation_Guest_id_Guest`, `Reservation_Guest_Barcode_id_Barcode`,  
  `Reservation_Guest_Barcode_Room_id_Room`)  
    ON DELETE NO ACTION  
    ON UPDATE NO ACTION)  
ENGINE = InnoDB;
```

Chapter Four: Implementation and Testing

4.1 Layouts

The Layouts are divided according to the user i.e. the guest interface, service provider interface, and admin interface which will be explained in this section.

4.1.1 Guest

Figure 18 represents the first interface of the guest in the application, here the guest enters the mobile number and then presses send, then the verification code comes on the mobile number and puts the numbers in the verification code. The main interface after logging in represented in Figure 19. There is the guest's name, personal photo, the notification sign, and the barcode assigned to his room with its location, as well as the time of entering and exiting the room. As it is shown in Figure 20, the interface for ordering room cleaning where the guest chooses the appropriate time for cleaning.



Figure 18. SignIn (Guest)



Figure 19. Guest Homepage



Figure 20. Room Cleaning

Figure 21 represents the profile interface to modify the data. Figure 22 represents the interface to chat with employees for feedback and inquiries.



Figure 21.Profile (Guest)



Figure 22.Chat (Guest)

Figure 23 represents the notification interface for the guest to know that the cleaning has been completed and others.



Figure 23. Notification (Guest)

4.1.2 Service Provider

Figure 24 represents the services interface for the service provider to know the requests of the guests over time and the status of the service, whether it has been completed or not, If the color of the case is green, the application has been completed, if the color is yellow, the application is being executed, and if the color is red, the application has not been initiated.



Figure 24. Service provider interface

In figure 25, when entering the new request, the service status changes from red to yellow. The service provider goes to the requested room and shows the barcode at the same time as the service. After completing the service, the service provider presses the finish sign and then send.

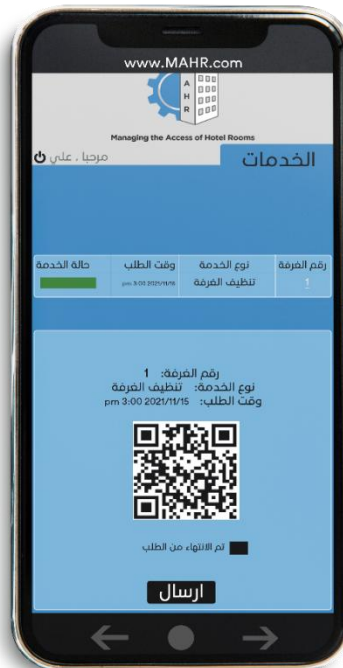


Figure 25. Service delivery

4.1.3 Admin

Figure 26 represents the login interface for the manager is to be logged in via e-mail and password.

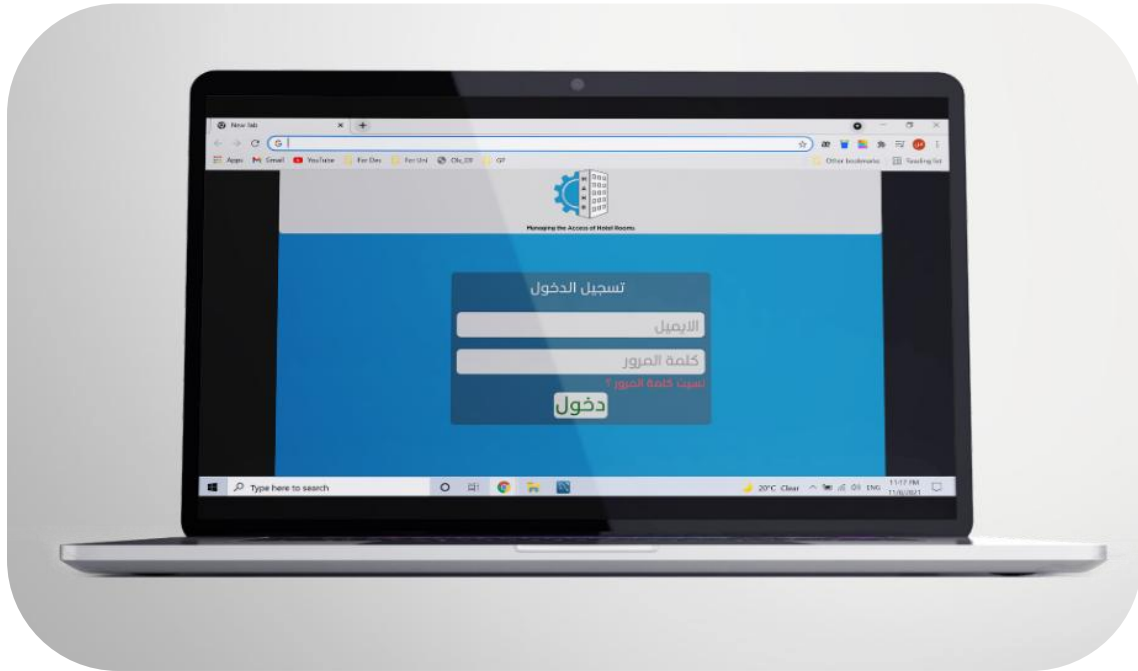


Figure 26. Log in (Admin)

Figure 27 represents the manager's page where they see the status of services for guests to ensure the progress of work as well as a chat icon which enable the manager to communicate with guests.

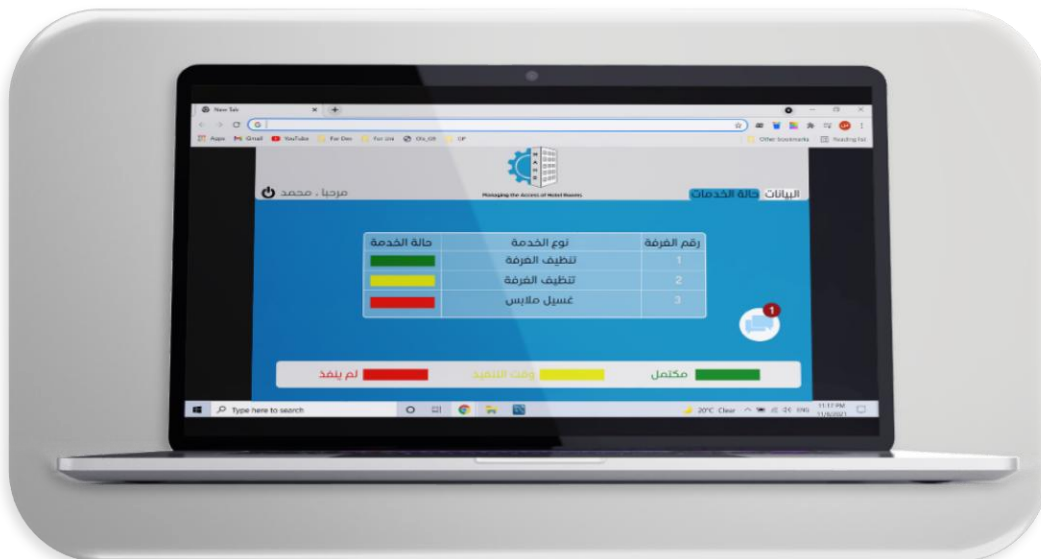


Figure 27. Service status

Figure 28 represent the layout when the user clicks on the chat sign, the manager will be able to communicate with the guests to know their inquiries and comments.

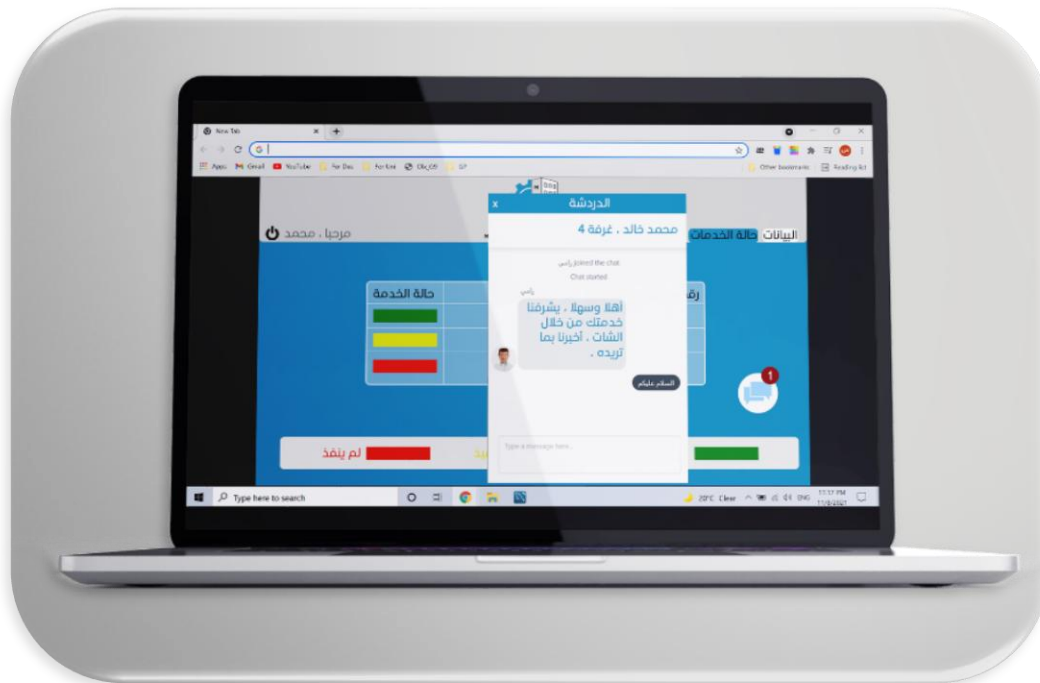


Figure 28.Chat

In figure 29, 30, 31 when the user clicks on the list of data, 3 lists of guests, reservations, employees will appear, in each list the manager can modify some fields, add, and delete.

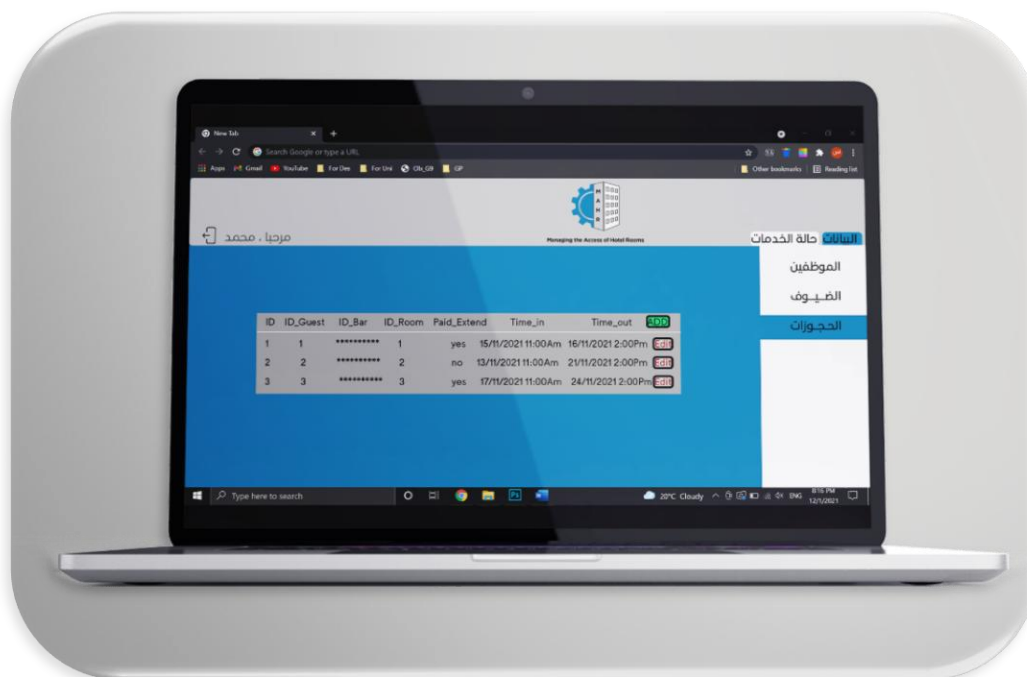


Figure 29.Data Management (Reservations)

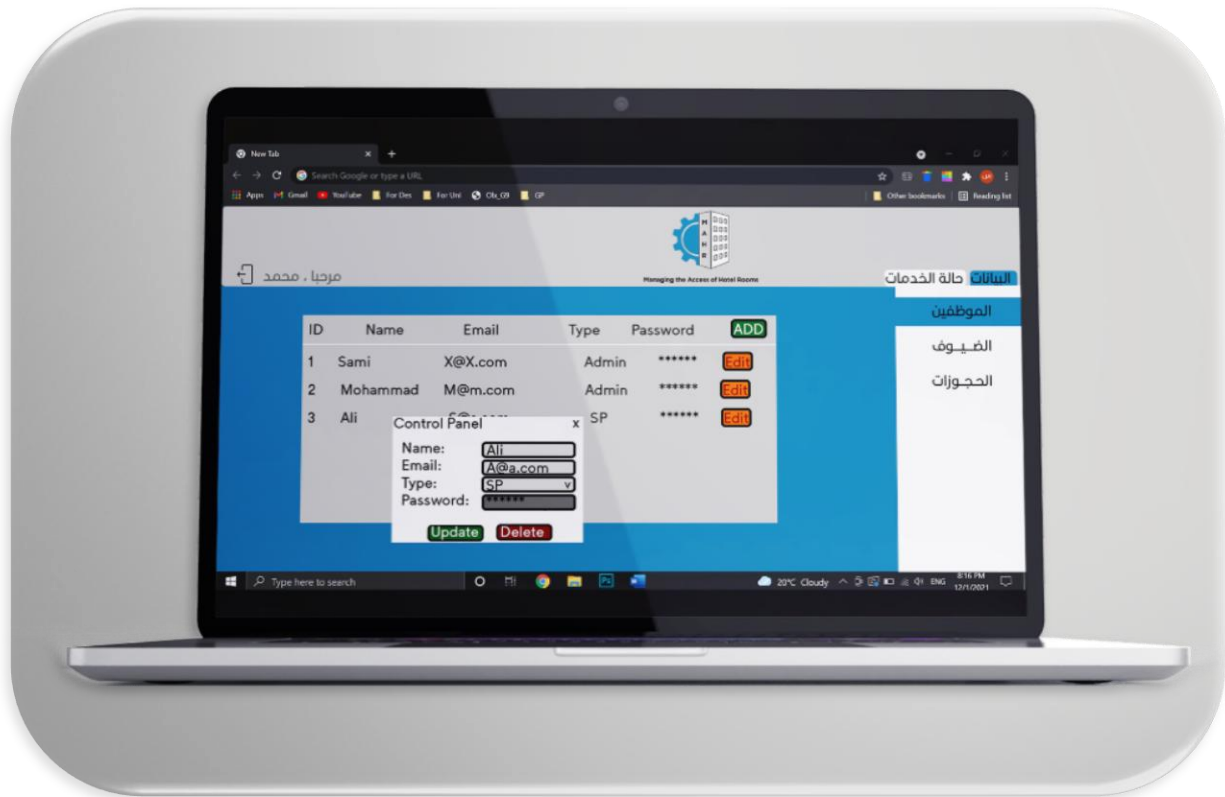


Figure 31.Data Management (Employees)

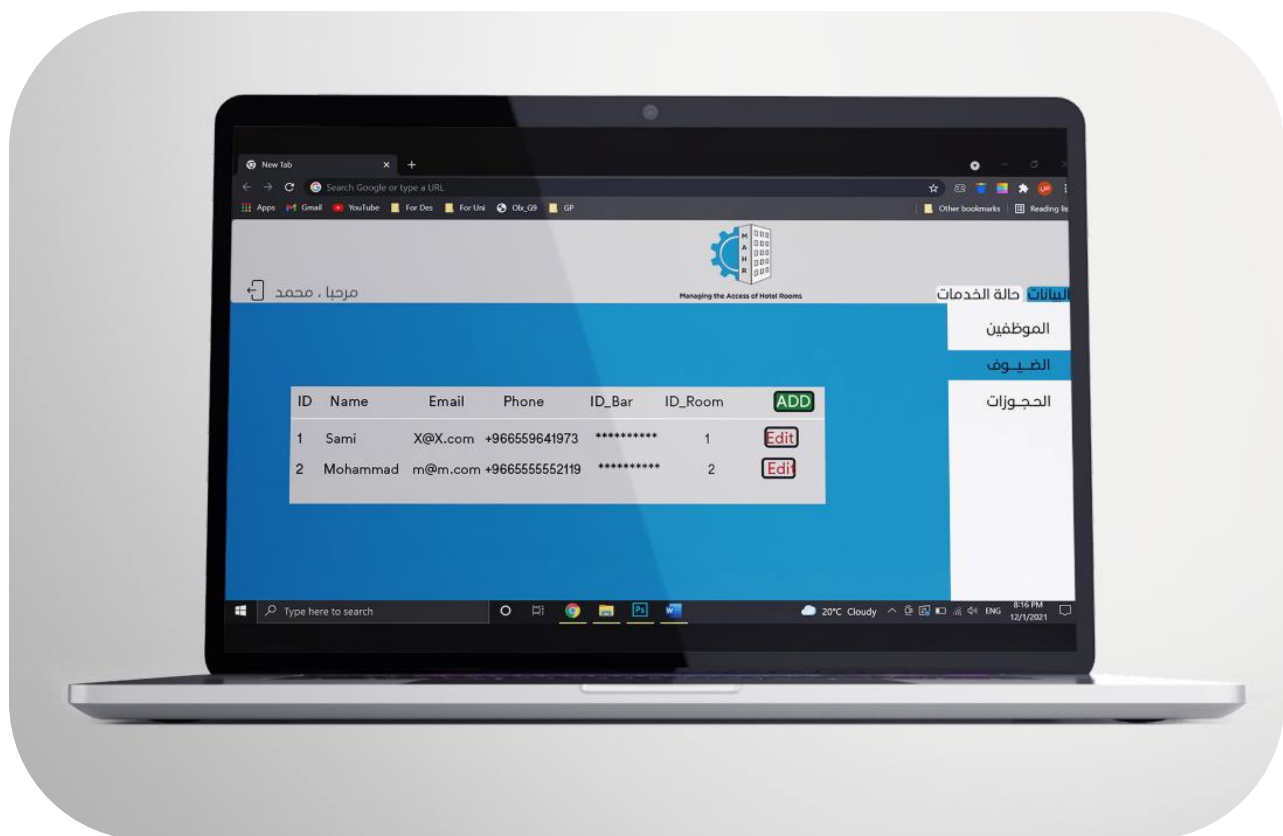


Figure 30.Data Management (Guest)

4.2 Conclusion

This project attempts to replace the traditional way of managing hotel access by a barcode-based system to facilitate the guests experience and hotel management. It addresses the problem of excessive consumption of card printing. To solve this problem, an app and a web browser are created to enable users to manage their reservation easily and quickly.

References

- Alvin, C., Peterson, B. & Mukhopadhyay, S. (2021) Static generation of UML sequence diagrams. *Int J Softw Tools Technol Transfer* 23, 31–53. <https://doi-org.sdl.idm.oclc.org/10.1007/s10009-019-00545-z>
- Bittner, K., Spence, I. (2003). *Use Case Modeling*, Addison Wesley.
- Open Courses. (2016, April 25). *Software Engineering |C5-L6| Sequence Diagram*. [video]. Youtube. <https://youtu.be/liwRAnMTVe8>
- Raval, K. M., Trivedi, S. H. (2013). A Study on Oracle Data Constraints. *International Journal*, 1(1). Retrieved from: https://www.academia.edu/4117875/A_Study_on_Oracle_Data_Constraints
- Shoval, P., Danoch, R., & Balabam, M. (2004). Hierarchical entity-relationship diagrams: the model, method of creation and experimental evaluation. *Requirements Engineering*, 9(4), 217–228. <https://doi-org.sdl.idm.oclc.org/10.1007/s00766-004-0201-9>
- Udacity. (2015, February 23). *Class Diagram: Relationships- Georgia Tech- Software Development Process*. [video]. Youtube. https://www.youtube.com/watch?v=E3IEGoXEQUM&ab_channel=Udacity
- Watt, A., Eng, N., (2014). Database design. Retrieved from: <https://openlibrary-repo.ecampusontario.ca/jspui/handle/123456789/247>